

Research papers

Unleashing the power of machine learning and remote sensing for robust seasonal drought monitoring: A stacking ensemble approach

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ABSTRACT

Droughts cause significant economic losses in many regions around the world, highlighting a need to more accurately quantify implications of drought on production and water management. Remote sensing technologies and machine learning-based models offer promising solutions for timely and accurate regional drought monitoring, but the accuracy of such approaches is constrained by both the conceptual design and algorithms underpinning such approaches. We developed a machine learning stacking ensemble approach to overcome such limitations, drawing upon Precipitation Estimation from Remotely Sensed Information using Artificial Neural Network-Climate Data Record (PERSIANN-CDR), Moderate Resolution Imaging Spectroradiometer (MODIS) remote sensing products, and climate zoning data to estimate 3-month scale SPEI (Standardized Precipitation Evapotranspiration Index, SPEI-3) in 9 sub-regions of China. We compared 19 individual machine learning models and used the stacking approach to select the most robust *meta*-model for SPEI-3 prediction. We found that CatBoost Regressor (CBR), Extra Trees Regressor (ETR), Extreme Gradient Boosting (XGB), Light Gradient Boosting Machine (LGBM), and Random Forest (RF) were the top individual models for predicting drought, while using CBR as the stacked *meta*-model achieved the best performance. The R^2 values for the stacking model with CBR as the *meta*-model were 0.9065 and 0.8218 in the eastern and western regions, respectively. We then employed the stacking model with CBR as the *meta*-model to generate seasonal drought maps based on SPEI-3 across different years and seasons. Our predicted SPEI-3 was compared with drought maps generated by the GPCC and ERA5 reanalysis datasets. We found a strong correlation between them with R^2 exceeding 0.8 across multiple years and seasons, indicating our machine learning stacking ensemble approach had a good performance in monitoring seasonal drought conditions. The findings of our study establish a standardized protocol for predicting droughts in the agricultural sector, enabling its widespread application irrespective of variations in terrain and climate.

1. Introduction

Drought impacts a significant proportion of the global population each year (Batista et al., 2015; Lobell and Gourdji, 2012; Wang et al., 2023). Drought may cause substantial losses in production due to either reduced cropping areas and/or reduced yields (Corey et al., 2016). Chronic and isolated droughts together cause annual global economic losses of \$US6-8B, making the phenomenon the most costly meteorological disaster facing humanity (Wilhite, 2000). The frequency and

intensity of drought disasters are increasing, as evident from occurrence of mega-drought events and anomalous performance characteristics (Zhang et al., 2020). Hence, the development of accurate and timely drought monitoring models is crucial for ensuring and promoting agricultural production, as well as facilitating disaster prevention and reduction efforts.

Climatic indices are crucial metrics in the objective measurement and contrasting of the severity and length of droughts, as they simplify intricate connections between numerous climate variables measuring

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and contrasting the severity and length of droughts (Alam et al., 2017). However, due to interacting factors and manifestations of drought, a single drought index cannot fully reflect its entire picture. Indeed, perceived 'drought' in agronomic contexts may depend on occurrence of extreme temperatures, rainfall or soil water deficit, extreme vapour pressure deficit, or combination thereof (Ibrahim et al., 2019). To address this limitation, there have been proposals for over 100 drought indices to characterize drought (Amir et al., 2011). Among these indices, the Standardized Precipitation Index (SPI) is a commonly used metric for measuring precipitation anomalies over various time frames (Mckee et al., 1993), which indicate how abnormal current precipitation patterns are compared to historical records at a specific location. Nevertheless, the SPI only focuses on the precipitation aspect of drought and disregards the impacts of temperature on water consumption and evapotranspiration. To overcome this limitation, the Standardized Precipitation Evapotranspiration Index (SPEI) was developed to integrate both precipitation and evaporation data (Vicente-Serrano et al., 2010). SPI is only suitable for assessing short-term drought, while SPEI can assess drought conditions at different time scales, which is important for the evaluation of seasonal drought (Pei et al., 2020). SPEI has been widely used to analyze the impact of drought on crops in various regions, including China (Ming et al., 2015), the United States (Moorhead et al., 2015), Australia (Feng et al., 2019), and Europe (Gunst et al., 2015), and has demonstrated effective results in evaluating the impact of meteorological drought on agricultural production (Yan et al., 2022).

Historically, drought monitoring has focused on station-based data because station meteorological data can provide more precise and reliable meteorological parameters, to accurately calculate drought indices (West et al., 2019). However, this approach is limited due to the sparse distribution of meteorological observations and the heterogeneity of underlying surfaces, and difficulty accessing data in many nations (Liu et al., 2023), leading to inaccurate regional assessments of water conditions (Cai et al., 2010). To overcome these challenges, some scholars employ spatial interpolation techniques via geostatistics to acquire information on regional drought conditions by utilizing meteorological data collected from various weather stations (Afzali et al., 2016; Akhtari et al., 2009; Eivazi and Mosaedi, 2011). However, due to the insufficient availability of meteorological observations and a lack of consideration for local climatic and topographical differences, this method is considered unreliable, impeding accurate assessment of drought conditions (Easterling, 2013).

Remote sensing enables seamless monitoring of the Earth's drought status by providing high-resolution and continuous spatiotemporal observations (Yao et al., 2023). Some studies have used remote sensing data or reanalysis data with similar characteristics to remote sensing data for regional drought analysis (Chen et al., 2019b; Ma et al., 2023; Zhou et al., 2020). However, the accuracy of both remotely sensed and reanalysis data is subject to factors such as observational data and numerical model biases, which may result in certain errors and uncertainties. Moreover, reliable validation of remotely sensed and reanalysis data still presents challenges, which could potentially impact the correctness and application value of research conclusions.

Currently, machine learning is rapidly advancing and has broad applications (Prodhan et al., 2022). In the context of drought monitoring, machine learning can leverage various techniques such as data fusion, feature selection, pre-processing, data harvesting, and classification to improve the accuracy and reliability of drought indices, providing scientific basis and support for drought prevention and control efforts. These techniques help extract valuable insights from multiple data sources, generate precise drought indicators, and play critical roles in predicting and managing droughts (Felsche and Ludwig, 2021). Currently, there are already numerous data fusion-based methods for drought monitoring and prediction (Alizadeh and Nikoo, 2018; Arabzadeh et al., 2015; Jia et al., 2022; Shahbazi et al., 2012; Yang et al., 2021). Many studies have also employed machine learning methods to characterize and predict drought, in order to investigate the impacts of

drought on agricultural production (Cai et al., 2023; Feng et al., 2019; Hazaymeh and Hassan, 2017; Marj and Meijerink, 2011; Woli et al., 2013). However, machine learning algorithms are still limited in terms of generalizability and stability. Recently, ensemble machine learning methods have been developed to exploit the benefits of multiple models (Tao et al., 2023). For example, stacking-based ensemble learning models combine the strengths of multiple models to enhance accuracy and stability (He et al., 2021). Wang (2022) combined multiple individual machine learning models with ensemble learning algorithms and applied the stacking ensemble learning architecture to soil moisture inversion, which significantly improved the model accuracy. These ensemble models outperform traditional mathematical models and methods in monitoring and predicting drought disasters (Prodhan et al., 2022) because they better account for variability across sites, climates and production systems (Sándor et al., 2020). Nevertheless, there have been few studies applying the stacking ensemble learning method to monitor seasonal drought, much less studies combining machine learning methods with remote sensing and climatic zonation approaches.

Drought disasters are common in China, with severe droughts occurring on average every 2.7 years and extreme droughts occurring on average every 5.5 years (Abrar Faiz et al., 2022; Ni et al., 2022). Frequent droughts in China have significantly impacted agriculture, causing substantial crop yield losses. Therefore, better monitoring of droughts to allow timely warnings will be imperative to enhance the China's capacity and effectiveness in responding to seasonal droughts. The purpose of this study is to use an ensemble machine learning method to integrate remote sensing drought factors for monitoring seasonal droughts. Specifically, our aims were to: 1) quantify the accuracy of 19 machine learning models in drought inversion; 2) assess the performance of 18 models as meta-models integrating 5 optimal individual models to build a stacking model; 3) evaluate the results in cropping region with SPEI-3 estimated from GPCC and ERA5 reanalysis datasets.

2. Materials and methods

2.1. Study area

China possesses a rich and diverse set of natural geographical elements with significant regional climate variations, as evidenced by climate models depicting notable regional heterogeneity. (Chen et al., 2019a). According to the "Nine Agricultural Regions of China" provided by Resource and Environmental Science and Data Center, China is divided into nine agricultural regions based on factors such as climate, land use, and crop types (Fig. 1a). The agricultural zoning data, climate zoning vector data, and cropland raster data based on China's land use map used in this study were obtained from the Resource and Environment Science and Data Center (RESDC) of the Chinese Academy of Sciences (<https://www.resdc.cn>). Furthermore, the final model inversion of drought conditions will be presented in China's cropping region (Fig. 1c) to provide drought indicators for agricultural production. The specific data for each subregion can be found in Table S1 of the Supplementary Materials document.

2.2. Data collection

The data used in this study included 762 station records, China's climate zoning data, China's land use data for the year 2010, Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks - Climate Data Record (PERSIANN-CDR), and Moderate Resolution Imaging Spectroradiometer (MODIS) satellite sensor data. Global Precipitation Climatology Centre (GPCC) products and Fifth generation of ECMWF atmospheric reanalyses of the global climate (ERA5) were also utilized for result comparison. The study period is from 2001 to 2017, with a data scale of 3 months and a time resolution of 1 month. Therefore, all data were merged at a 3-month scale, and the

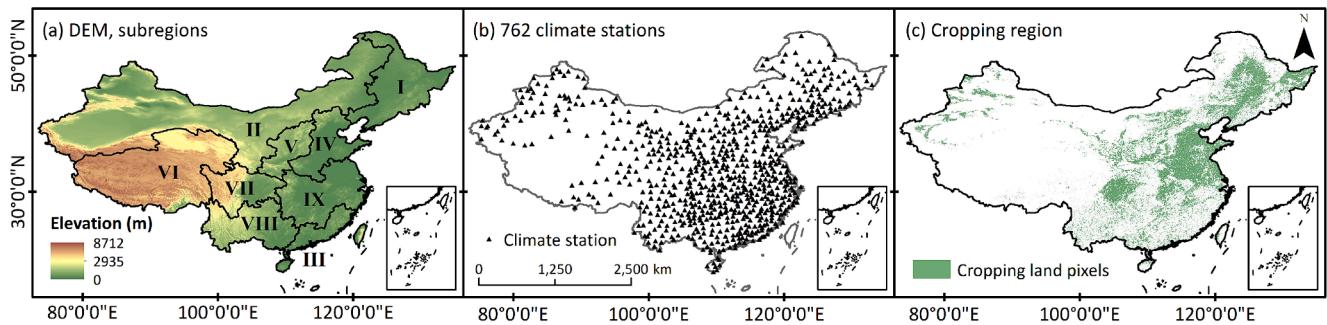


Fig. 1. Spatial distribution of (a) digital elevation and 9 sub-regions, (b) 762 climate stations, and (c) the cropping region across China's mainland in 2010. I: Northeast Plain; II: The Semi-Arid and Arid Areas of Northern China; III: South China Region; IV: Huang-Huai-Hai Plain; V: Loess Plateau Region; VI: Qinghai-Tibet Plateau; VII: Sichuan Basin and Its Surrounding Areas; VIII: Yunnan-Guizhou Plateau; and IX: The Middle and Lower Reaches of the Yangtze River.

remote sensing data from the last two months of 2000 were also included. Table S2 of the Supplementary Materials document displays the time span and source of the data.

2.2.1. Ground-based climate data

The climate station observation data used in this study were obtained from the China Meteorological Data Service Center (<https://data.cma.cn/>). Daily data for 762 stations including precipitation, maximum and minimum temperatures, and radiation from 1961 to 2017 were used to derive SPEI at the station level. Fig. 1b shows the distribution of climate stations across mainland China.

2.2.2. Remote sensing data

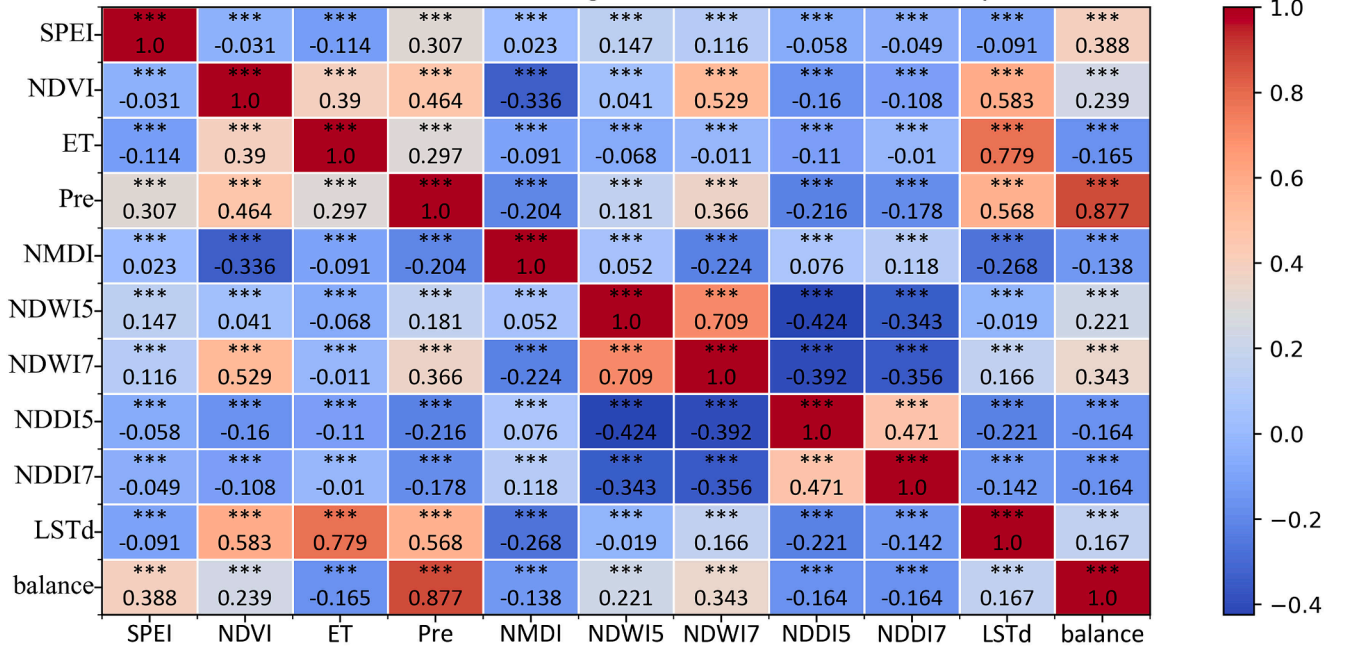
The remote sensing data used drought prediction in this study were obtained from Google Earth Engine (GEE, <https://earthengine.google.com>). The Evapotranspiration (ET) and Potential Evapotranspiration (PET) data were derived from the MODIS MOD16A2.006: Terra Net Evapotranspiration 8-Day Global 500 m dataset. The ET and PET data in this dataset were calculated using the Penman-Monteith equation, incorporating MODIS remote sensing data products and meteorological reanalysis data, including radiation, temperature, humidity, wind speed, as well as vegetation characteristics and land cover data products. The Land Surface Temperature (LST) data were obtained from the MODIS MOD11A1.061 Terra Land Surface Temperature and Emissivity Daily Global 1 km dataset. Precipitation, evaporation, and temperature are directly linked to the water balance within a region. By incorporating these factors, the model can capture the essential elements that contribute to drought conditions. If certain variables are less important for predicting SPEI, they will rank lower in the final importance ranking. The surface reflectance data were acquired from MODIS MOD09A1.006 Terra Surface Reflectance 8-Day Global 500 m dataset. NDVI (Normalized Difference Vegetation Index), NMDI (Normalized Multi-band Drought Index), NDWI (Normalized Difference Water Index), and NDDI (Normalized Difference Drought Index) indices were calculated based on the surface reflectance of seven bands (B1-B7) of the dataset (Table 1). Vegetation indices like NDVI are sensitive to changes in vegetation health and density. They can capture variations in vegetation growth and vigor, which are influenced by water availability. NDWI measures the difference between near-infrared and shortwave infrared bands, providing information on the presence of water bodies, moisture content in vegetation, and soil moisture. NMDI focuses on the normalized difference between near-infrared and mid-infrared bands, indicating the level of drought stress in vegetation. By including these indices as variables, we can better understand the impact of vegetation on drought conditions. The remote sensing datasets we used have undergone certain preprocessing (e.g. cloud removal), and only require a resampling operation to standardize the resolution to 1 km. The resampling method used is the built-in bilinear interpolation provided by Google Earth Engine.

Table 1
Remote sensing indices for estimating seasonal drought and evaluating the results.

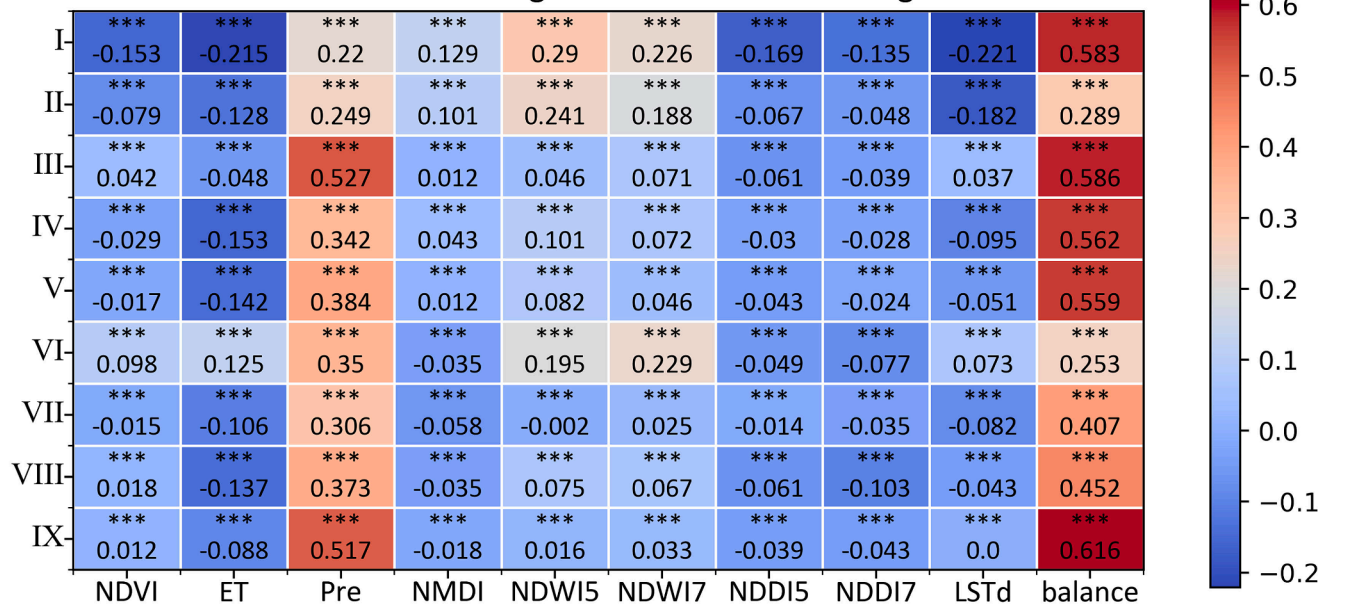
Indices	Definition	Formula	Resolution	Source
Pre	Precipitation	–	0.25°	GPCC
		–	0.25°	ECMWF
		–	0.25°	ERA5
		–	0.25°	NOAA
PET	Potential evapotranspiration	–	500 m	PERSIANN-CDR MODIS
ET	Evapotranspiration	–	500 m	MODIS
LSTd	Land surface temperature-day	–	1 km	MODIS
NDVI	Normalized difference vegetation index	$(b2-b1)/(b2+b1)$	500 m	Rouse et al. (1974)
NMDI	Normalized multi-band drought index	$(b2-(b6-b7))/(b2+(b6-b7))$	500 m	Wang and Qu (2007)
NDWI5	Normalized difference water index	$(b2-b5)/(b2+b5)$	500 m	Gao (1996)
NDWI7	Normalized difference water index	$(b2-b7)/(b2+b7)$	500 m	Gao (1996)
NDDI5	Normalized difference drought index	$(NDVI-NDWI5)/(NDVI+NDWI5)$	500 m	Gu et al. (2007)
NDDI7	Normalized difference drought index	$(NDVI-NDWI7)/(NDVI+NDWI7)$	500 m	Gu et al. (2007)

The precipitation (Pre) data were obtained from the NOAA PERSIANN-CDR dataset, with a spatial resolution of 0.25 degrees and a temporal resolution of 1 day. Furthermore, given that SPEI considers the effects of both precipitation and potential evapotranspiration, we introduced a simple water balance index called ‘balance’ (Pre - PET), which represents water balance status in a certain period. To avoid multicollinearity and the curse of dimensionality, we calculated the variance inflation factor (VIF) for each variable and further eliminated variables with high VIF values and relatively low importance (Robert, 2007). This process improves the performance and interpretability of machine learning. Fig. 2a shows the Pearson correlation coefficient between SPEI and retained predict variables. It shows that selecting balance as a feature variable is reasonable, as it exhibits the highest correlation with SPEI among all the examined features, with a Pearson correlation coefficient of 0.388. Fig. 2b shows the Pearson correlation coefficient between the SPEI and each drought factor across 9 sub-regions, revealing that the dominant factors influencing the SPEI of each sub-region vary.

(a) The Pearson correlation coefficient among the 3-month scale SPEI and 10 drought factors used in this study



(b) The Pearson correlation coefficient between 3-month scale SPEI and each drought factor across 9 sub-regions



III: South China Region

V: Loess Plateau Region

VII: Sichuan Basin and Its Surrounding Areas

IX: The Middle and Lower Reaches of the Yangtze River.

IV: Huang-Huai-Hai Plain

VI: Qinghai-Tibet Plateau

VIII: Yunnan-Guizhou Plateau

*** significant level (p-value<0.001)

Fig. 2. The Pearson correlation coefficient among the 3-month scale SPEI and 10 drought factors used in this study (a), and the Pearson correlation coefficient between 3-month scale SPEI and each drought factor across 9 sub-regions (b). The definition of remote sensing drought factors can be found in Table 1. The values in each cell represent the correlation coefficient between the corresponding pair of variables.

2.2.3. Reanalysis data

We selected the GPCC reanalysis, which is generated based on global precipitation observational data and reanalysis methods, and the ERA5 dataset, which is generated through the fusion of numerical models and observational data, as our verification data. By using these two different

data processing methods, we can conduct a more comprehensive validation and comparison of our research results, thereby increasing the reliability of the data. These datasets have been proven to have high accuracy and are widely used for verification and comparison in drought assessment (Chen et al., 2019b; Ma et al., 2023; Wei et al., 2023). The

GPCC precipitation dataset is based on approximately 85,000 observation points worldwide and utilizes the SphereMap interpolation method to derive a global gridded land precipitation dataset (Becker et al., 2013). The latest GPCC version 2022 product is accessible via https://opendata.dwd.de/climate_environment/GPCC/html/download_gate.html. ERA5 is the fifth generation of reanalysis products developed by ECMWF, which is used to reanalyze global atmospheric changes using models and data. It couples data from diverse sources, including satellites, meteorological observation stations, and aircraft, with advanced physical models to calculate high-precision spatiotemporal distribution information on various meteorological variables globally (Hersbach et al., 2020). The dataset covers data from 1979 to the present and can be accessed via <https://cds.climate.copernicus.eu/>.

2.2.4. Climate zoning data

Due to the substantial differences in geography and climate across different regions of China, we divided the whole country into 46 climatic regions to establish a geographically targeted model that improves the accuracy of predictions (Yin, 2017). The data on climatic zoning, as a qualitative variable, is transformed into a binary variable recognizable by machine learning algorithms through dummy variable processing, which participates in machine learning analysis. Please refer to Fig. S1 of the Supplementary Materials document for the map of climatic regions

and corresponding names of the regions.

2.3. Methods

The flowchart shown in Fig. 3 illustrates the overall process of the stacking-model-based methodology used for estimating seasonal drought. The process begins with the collection of predictor variables from remote sensing data sources such as MODIS and PERSIANN-CDR, which include variables like ET, PET, LST, and Reflectance, and precipitation data, along with climate zoning data that has been transformed using dummy variable encoding. These predictors feed into the machine learning models for drought estimation. In this study, an advanced model ensemble approach based on stacking was utilized to combine the top 5 individual machine learning models for each agricultural region, using these predictors to estimate SPEI at a 3-month scale for analyzing and estimating seasonal drought. The response variable is the SPEI calculated from data gathered at 762 climate stations, which serves as the ground truth for model validation. First, we selected the top 5 machine learning models in estimating drought from 19 algorithms using the aforementioned predictors for each subregion. Second, we evaluated the performance of 18 meta-models (excluding KNR) in stacking ensembles and subsequently selected the best-performing one to integrate with the top 5 individual models. Stacking typically involves combining and adjusting base model predictions using a meta-

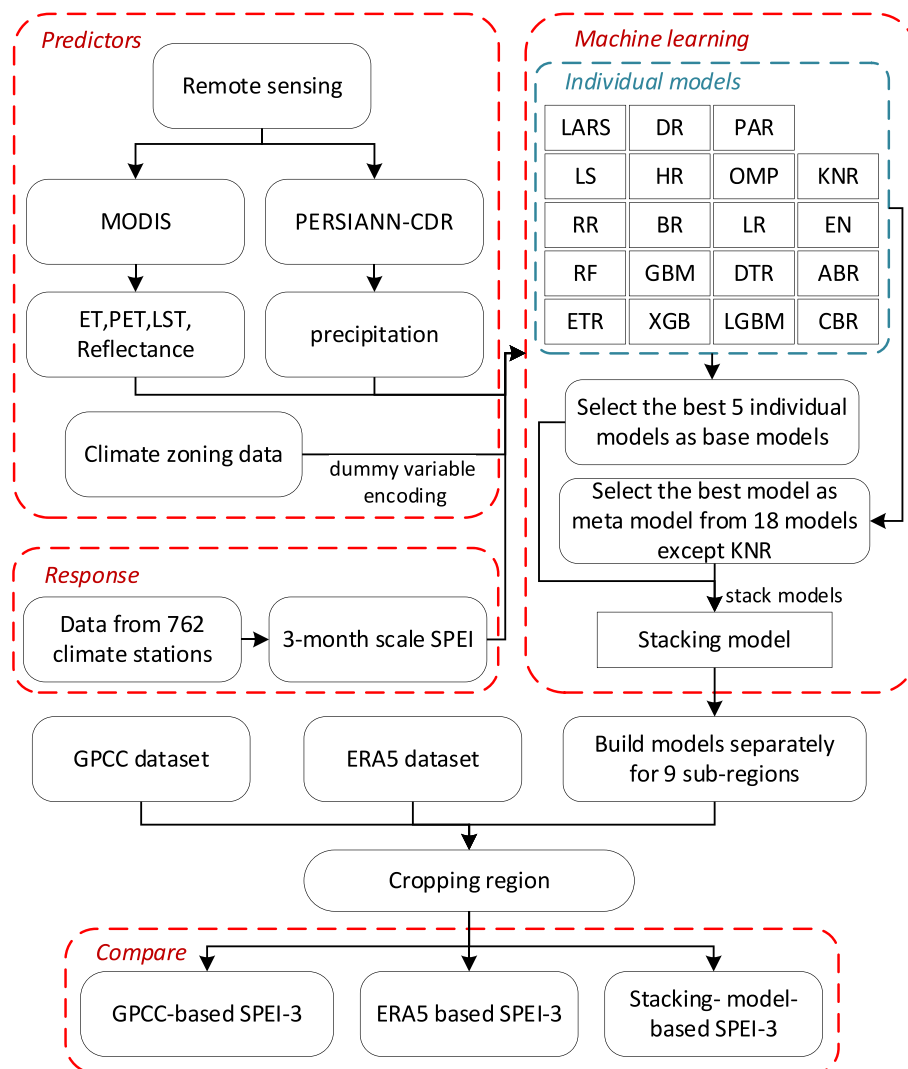


Fig. 3. Flowchart of the proposed model stacking methodology based on multiple data sources for estimating drought.

model, but as KNR only relies on neighboring data points without modeling complex relationships, we excluded it from our *meta-model* comparison. The subsequent sections will provide a detailed description of the principal components of the proposed procedures.

2.3.1. Standardised precipitation evapotranspiration index (SPEI)

SPEI utilizes periodic differences between precipitation and PET (Vicente-Serrano et al., 2010). Therefore, we first used the Penman-Monteith method to calculate the monthly potential evapotranspiration because it incorporates the effects of temperature, radiation and other crucial factors (Trenberth et al., 2014).

Then, we calculate the difference between monthly precipitation and potential evapotranspiration, fit this data using a three-parameter log-logistic distribution, obtain its cumulative function, standardize the sequence through normal processing, and ultimately acquire the SPEI (Vicente-Serrano et al., 2010). The drought levels corresponding to different SPEI values are categorized into severe, moderate, and mild drought for SPEI values of -2 or below, between -1.5 and -2 , and between -1.5 and -1 , respectively, while SPEI values between -1 and 1 indicate normal conditions that are not affected by drought or moisture.

2.3.2. Machine learning models

The capability of individual machine learning models has a significant impact on the effectiveness of model ensemble. Therefore, we compared 19 commonly used machine learning models (Table 2). Since the parameter settings have minimal impact on the results, we used default parameters for all machine learning algorithms in all regions. We have provided a detailed introduction of the algorithms primarily used in this study in the [Supplementary Materials](#).

Catboost is a machine learning model based on gradient boosting decision trees (Dorogush et al., 2018). It shows exceptional performance when compared to other individual models, whether as an individual model or as a *meta-model* in the ensemble. Catboost effectively handles categorical features and minimizes information loss (Prokhorenkova et al., 2018). It uses a random permutation method to select tree structures for estimating leaf values, thus avoiding overfitting issues commonly encountered in gradient boosting algorithms (Ullah et al., 2022). The ability of Catboost to handle categorical features makes it particularly suitable for many real-world applications, such as the use of Catboost for soil water content based on multi-sensor data by Nguyen et al. (2022).

The 4 individual models, ETR, XGB, LGBM, and RF, are also used as base models in model ensemble due to their excellent performance. All of them are based on the ensemble method of decision trees. ETR is an ensemble learning method that utilizes randomization to produce a set

of decision trees and then averages their outputs. Compared to other ensemble methods, Extra Trees is simpler, more easily scalable, and can handle large amounts of data (Geurts et al., 2006). XGB is an efficient gradient boosting decision tree (GBDT) algorithm that employs multiple optimization techniques to accelerate the training process, including approximate greedy algorithms, distributed computing, and caching techniques, demonstrating outstanding accuracy and speed (Chen and Guestrin, 2016). LGBM is also a GBDT algorithm that utilizes histogram-based decision tree algorithms to discretize continuous value features, thereby significantly reducing memory consumption and calculation time (Ke et al., 2017). RF is a popular ensemble learning technique that employs multiple decision trees to produce prediction outcomes and aggregates them by either voting or averaging. Compared to single decision trees, Random Forest can reduce overfitting problems and is more robust in dealing with high-dimensional and missing data (Breiman, 2001).

2.3.3. Model stacking

Stacking is a commonly used method to produce an ensemble model where multiple base models' outputs are combined as features to train a *meta-model* for making final predictions on the target variable (Wolpert, 1992). In stacking, different base models and meta models can be selected to achieve better performance and generalization ability. The stacking model is actually the trained meta model, which we refer to as the meta model during the stacking process. Fig. 4 illustrates the principle of the stacking model that combines five models using a 5-fold cross-validation approach, which was employed in this study. The 5-fold cross-validation in the stacking model refers to the cross-validation of the training data set for each model run, which is the 'Training Sample Data' shown in Fig. 4.

In the first layer, there are 5 basic models, using *base model_i* as an example. During each model run, we had 90% of training data and 10% of testing data, denoted as 'Train' and 'Test' in the first layer, respectively. The details of how the data was split will be further explained in the 'Model performance assessment' section. The 'Train' data is randomly split into five parts for cross-validation. In the *j*th cross-validation iteration, we used four 'train' data to build a model, predict the 'validation data' to obtain a prediction value $a_{i,j}$, and then use the trained model to predict the 'Test' data to obtain a prediction value $b_{i,j}$. After 5-fold cross-validation, we got $\{a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}, a_{i,5}\}$ and $\{b_{i,1}, b_{i,2}, b_{i,3}, b_{i,4}, b_{i,5}\}$, stack the former vertically to form A_i with the same length as the training sample data, and take the average of the latter to obtain B_i with the same length as the testing sample data.

In the second layer, we used $\{A_1, A_2, \dots, A_5\}$, which are the outputs of 5 models in the first layer, as new training features. The label of 'Train' is used as training label to train a *meta-model*. Finally, the trained stacking model (*meta-model*) is used to predict label for $\{B_1, B_2, \dots, B_5\}$, which are the testing features. The predicted label is compared to the label of 'Test' to evaluate the stacking model's performance.

2.3.4. Model performance assessment

We constructed machine learning models using satellite-based remote sensing drought factors and ground-based observed SPEI-3 as predictors and response variable, respectively. We ensured the independence and balance between the training set and test set by employing *k*-fold cross-validation on the data. However, running the cross-validation process once may lead to inaccurate estimation of model performance, and different splits of the data can yield varying results. To address this, we conducted ten rounds of 10-fold cross-validation, resulting in a total of 100 experiments. In each round of 10-fold cross-validation, the dataset was divided into 10 parts, with 9 parts used as training data and 1 part as test data. When building the stacking model, we further performed 5-fold cross-validation on the training set (Fig. 4). By performing multiple cross-validations, we mitigated the impact of randomness and enhanced the reliability of the experimental results. It is worth mentioning that each experiment's training and testing data were

Table 2
The 19 individual machine learning models used in the study.

	Model	Source
ABR	AdaBoost Regressor	–
BR	Bayesian Ridge	–
CBR	CatBoost Regressor	Dorogush et al. (2018)
DR	Dummy Regressor	–
DTR	Decision Tree Regressor	–
EN	Elastic Net	–
ETR	Extra Trees Regressor	Geurts et al. (2006)
GBM	Gradient Boosting Regressor	Friedman (2001)
HR	Huber Regressor	–
KNR	K Neighbors Regressor	–
LARS	Lasso Least Angle Regression	–
LGBM	Light Gradient Boosting Machine	Ke et al. (2017)
LR	Linear Regression	–
LS	Lasso Regression	–
OMP	Orthogonal Matching Pursuit	–
PAR	Passive Aggressive Regressor	–
RF	Random Forest Regressor	Breiman (2001)
RR	Ridge Regression	–
XGB	Extreme Gradient Boosting	Chen and Guestrin (2016)

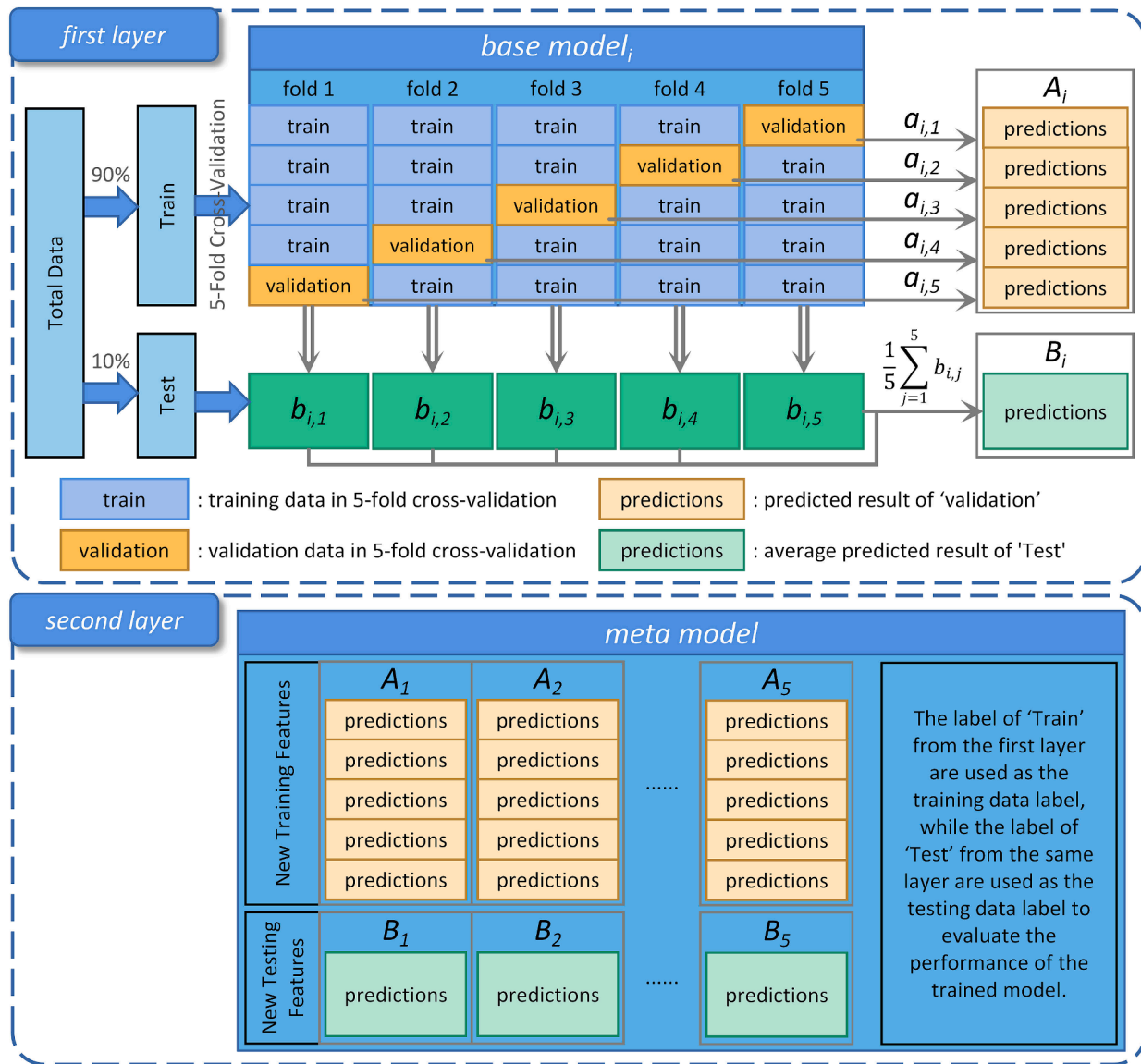


Fig. 4. Schematic representation of the proposed 5-fold cross-validation based stacking model approach.

independent. This approach is commonly used in machine learning to evaluate model performance (Kohavi, 1995).

In our study, we used multiple metrics to assess the performance of the models. We chose the coefficient of determination (R^2) as the primary metric because it reflects the model's ability to explain the variance of the target variable well. We also used mean squared error (MSE) as a secondary metric because it measures the magnitude of the error between predicted values and actual values. R^2 has a range of 0 to 1, with a higher value indicating a better ability of the model to explain the variance; MSE has a range from 0 to positive infinity, with a lower value indicating a better predictive capability of the model.

3. Results

3.1. Performance assessment of individual models

We performed ten 10-fold cross-validations on the data, and generated 100 training and validation data for model evaluation. After running each of 19 models for 100 times in 9 sub-regions, the average R^2 and MSE were obtained in Fig. 5. For all 9 sub-regions, CBR, ETR, XGB, LGBM, and RF showed the best performance, therefore we choose these

5 models as the base models for subsequent model ensemble.

3.2. Selection of the meta-model in stacking

When stacking models, the choice of the meta-model can directly affect the final fit. Fig. 6 shows the top 5 meta-models for each sub-region (The R^2 values for all meta-models can be found in Table S3 in the appendix, while the MSE values are in Table S4.). Due to its advantages of automatic handling of categorical features and strong robustness, CBR is the best meta-model in each sub-region. Therefore, CBR will be used as the meta-model for all subsequent stacking models.

3.3. Evaluation of the stacking model

We compared the performance of stacking versus 5 base models across 9 sub-regions, and the R^2 values of the results are shown in Fig. 7. The MSE values can be found in Appendix Fig. S2. Overall, the stacking model performed better than individual models in terms of both performance and stability. In the 5 regions located closer to the eastern part of China, including Northeast Plain, South China Region, Huang-Huai-Hai Plain, Loess Plateau Region, and The Middle and Lower Reaches

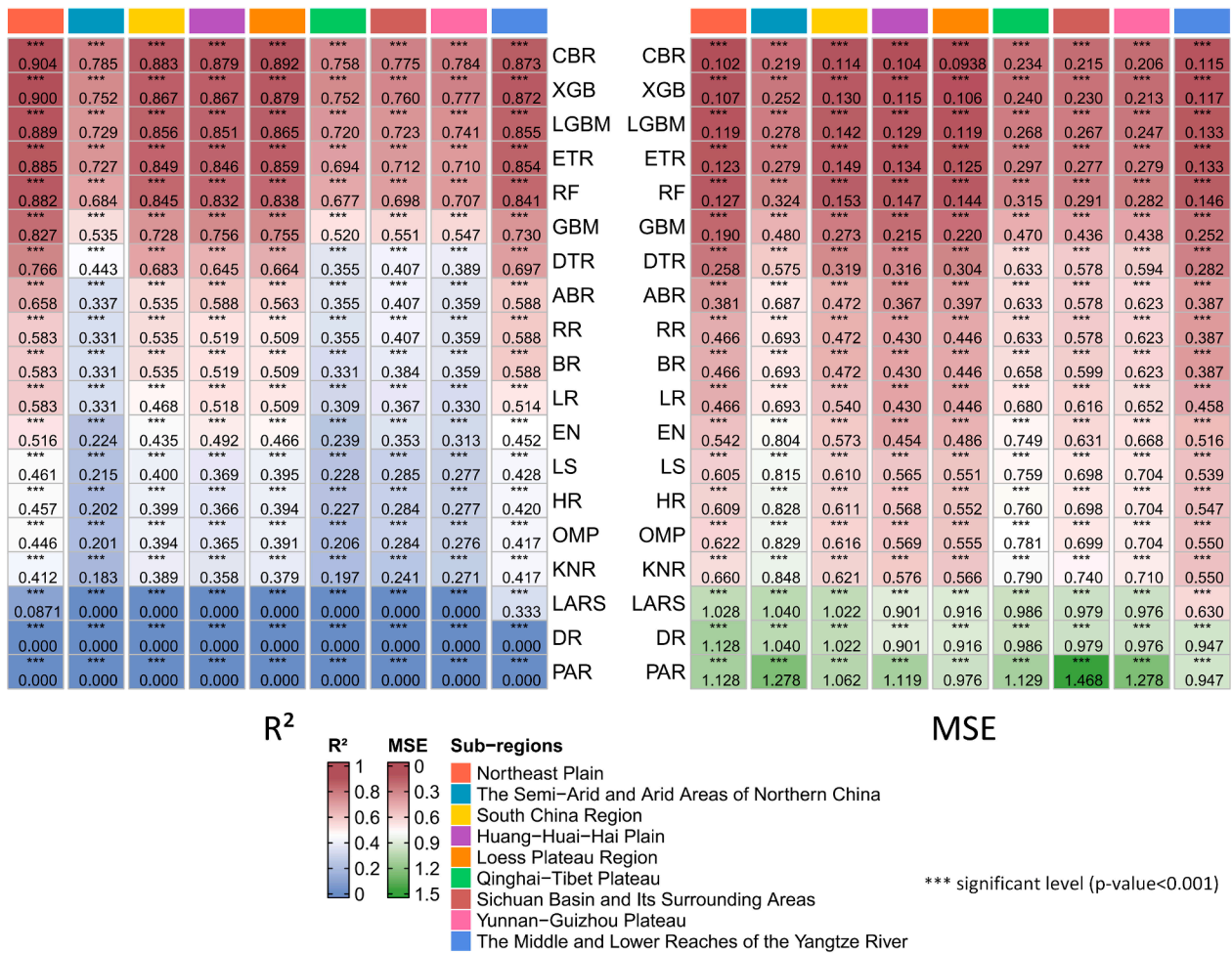


Fig. 5. The average of model performance measurements (R^2 and MSE) for 100 runs with 19 machine learning models in 9 sub-regions.

of the Yangtze River, the stacking model fitted R^2 values were all above 0.89, indicating a relatively excellent estimation performance. However, in the 4 regions located more towards the west, including the Qinghai-Tibet Plateau, Semi-Arid and Arid Areas of Northern China, Sichuan Basin and Its Surrounding Areas, and Yunnan-Guizhou Plateau, the stacking model's R^2 values were all above 0.8, which was marginally lower than that of the eastern regions. Combined with the station count (Table S1), increasing data has a small impact on the accuracy of machine learning estimates of drought, but it can improve the stability of the estimates to some extent.

We conducted 10-fold cross-validation on the stations within each sub-region to assess the performance of the stacking model in estimating droughts at individual points based on continuous time series. We chose to display three-month scale SPEI-3 for stations 54,063 and 57,411 from 2001 to 2017 because they are in the eastern and western regions, respectively, both situated in areas with abundant farmland clusters (Fig. 8). For images of the densely populated areas of other regions, please refer to Supplementary materials Figs. S3–S9. Fig. 9 presents the boxplot of R^2 values between observed and predicted SPEI-3 for the remaining stations in each subregion. Although the fitting effect of the station located in the eastern region was better than that of the western region, the station in the west was still able to clearly reflect the trend of drought occurrence.

3.4. Weighted relative importance of features in the stacking model

The weighted relative importance method of stacking model is to weight the predictions of each base model based on their score ratio in

cross-validation. Specifically, the weighted relative importance of each factor is the sum of its relative importance in each base model multiplied by the relative importance of the feature generated by the base model in the stacking model. For example, to calculate the weighted relative average of 'Pre', first calculate its relative importance in each base model, then calculate the relative importance of the feature corresponding to each model when constructing the stacking model. Multiply these two values for each model and sum the results to obtain the weighted relative average of 'Pre'. Fig. 10 shows the top 9 features with highest importance for each sub-region, while the importance scores of all features can be found in Table S5 in the appendix. In most regions, balance had the highest ranking, while in areas with scarce precipitation (The Semi-Arid and Arid Areas of Northern China and the Qinghai-Tibet Plateau), precipitation ranks highest. Climate zoning exerted a greater influence over larger areas. Meteorological factors generally had a greater contribution to model establishment compared to vegetation indices, while vegetation indices also played a role in improving the final prediction accuracy.

3.5. Comparison with reanalysis-based drought conditions

We conducted a comparative analysis of the stacking model-based SPEI, GPCC-based SPEI, and ERA5-based SPEI against observed values at the site scale, as illustrated in Fig. S10. Our findings reveal that the stacking model-based SPEI demonstrates superior accuracy and robustness relative to the other products. Subsequently, we utilized the stacking model to estimate drought conditions in the cropping region of China across various seasons and years, based on the SPEI-3. This

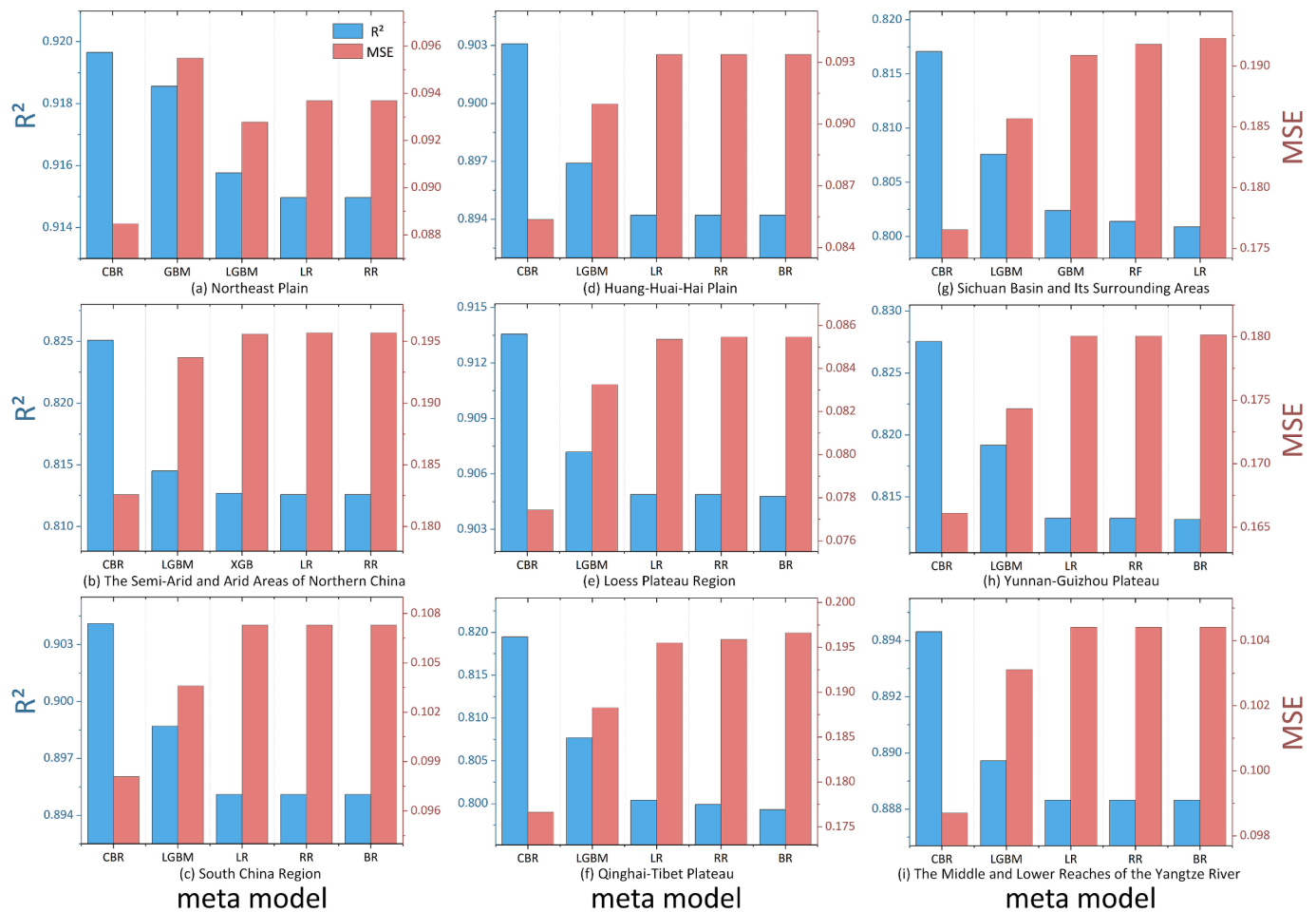


Fig. 6. The average R^2 and MSE based on 100 runs for top 5 performing meta-models selected from 18 models in 9 sub-regions.

estimation was then juxtaposed with results obtained from two reanalysis datasets, namely GPCC and ERA5, as depicted in Fig. 11. The SPEI-3 estimated by the stacking model exhibit a strong correlation with those from GPCC and ERA5, evidenced by R^2 values exceeding 0.8. The results indicate that the stacking model performs well in estimating large-scale drought conditions.

4. Discussion

Remote sensing products provide a wealth of spatial information and time series data, which makes it possible for accurate estimation of seasonal drought (Yin et al., 2022; Zuo et al., 2019). It is possible that there is a nonlinear relationship between SPEI and drought factors, and certain machine learning algorithms are well-suited for analyzing this type of relationship. In our study, we utilized the machine learning ensemble model approach along with remote sensing data to invert regional drought conditions. Our methods achieved robust results for seasonal drought estimation. Based on previous research (Feng et al., 2019; Kikon and Deka, 2022; Yang et al., 2021) and comparisons of numerous models in different regions, we found that the performance of machine learning methods for drought monitoring areas are region-specific. Of 19 machine learning models, CBR had the best performance in predicting SPEI as it reduces the need for tuning many hyperparameters and decreases the chance of overfitting, and its performance in estimating drought can rival that of most machine learning algorithms. Both Huang et al. (2019) and Katimbo et al. (2023) utilized CBR to estimate ET_0 and CWSI respectively, and CBR exhibited superior performance compared to other models in both cases.

The results of model comparison showed a great improvement in

prediction accuracy by integrating 5 models with the stacking method. However, there was a difference in prediction accuracy between the eastern and western regions of China, with the average R^2 of the stacking model being 0.91 in the 5 eastern regions and 0.82 in the western region. This difference in prediction accuracy can be attributed to a combination of factors, such as topography, humidity, data quality, monsoon, and drought frequency. Fig. 1a shows the topography of China, with sub-regions in the west located mostly on the first and second tiers of China with high elevation and undulating terrain, while sub-regions in the east are generally characterized by lower or flatter terrain. According to Table S1, the average annual precipitation in the 5 eastern regions is 922 mm, while that in the 4 western regions is only 441 mm. Therefore, the estimation performance will be better in humid regions, which is consistent with the conclusion of Feng et al. (2019). Furthermore, Ma et al. (2023) showed that multi-source precipitation estimate products, including satellite-based PERSIANN-CDR and GPCC/ERA5 reanalysis data, performed better in calculating SPEI in the eastern regions of China. Compared to the approach of using multi-source precipitation estimate products for drought estimation from Ma et al. (2023) and Chen et al. (2019b), we made a progress in terms of accuracy and incorporating ground-based observation data for further validation.

In addition, the importance of different remote sensing drought factors varies in different regions (Fig. 10). For example, the 'balance' index, which reflects the difference between precipitation and PET, dominates the importance in most regions. The importance of precipitation is relatively high in areas with less rainfall, while the importance of climate zoning increases in larger areas, such as The Semi-Arid and Arid Areas of Northern China and The Middle and Lower Reaches of the Yangtze River. Compared with meteorological indices, the importance

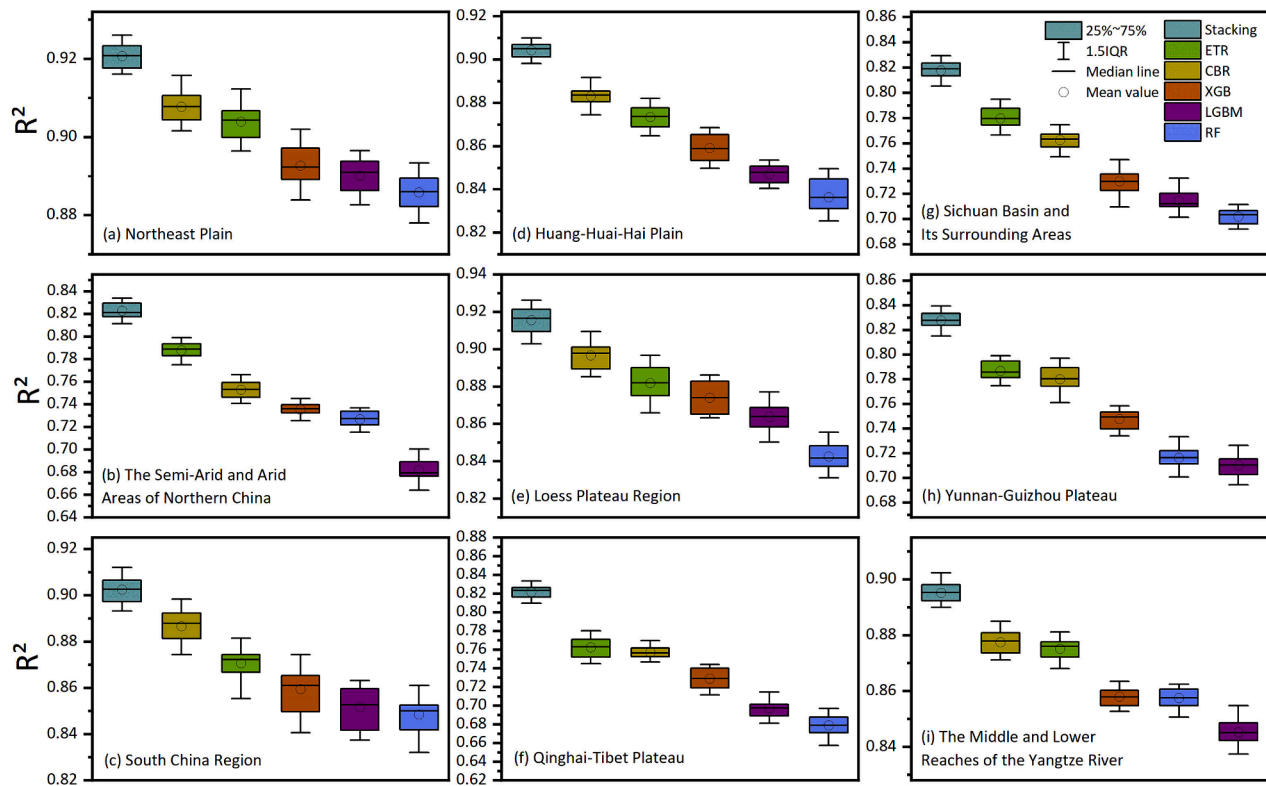


Fig. 7. Boxplots of the R^2 values derived from stacking models and 5 individual models (ETR, CBR, XGB, LGBM, and RF) across 9 sub-regions based on 100 runs.

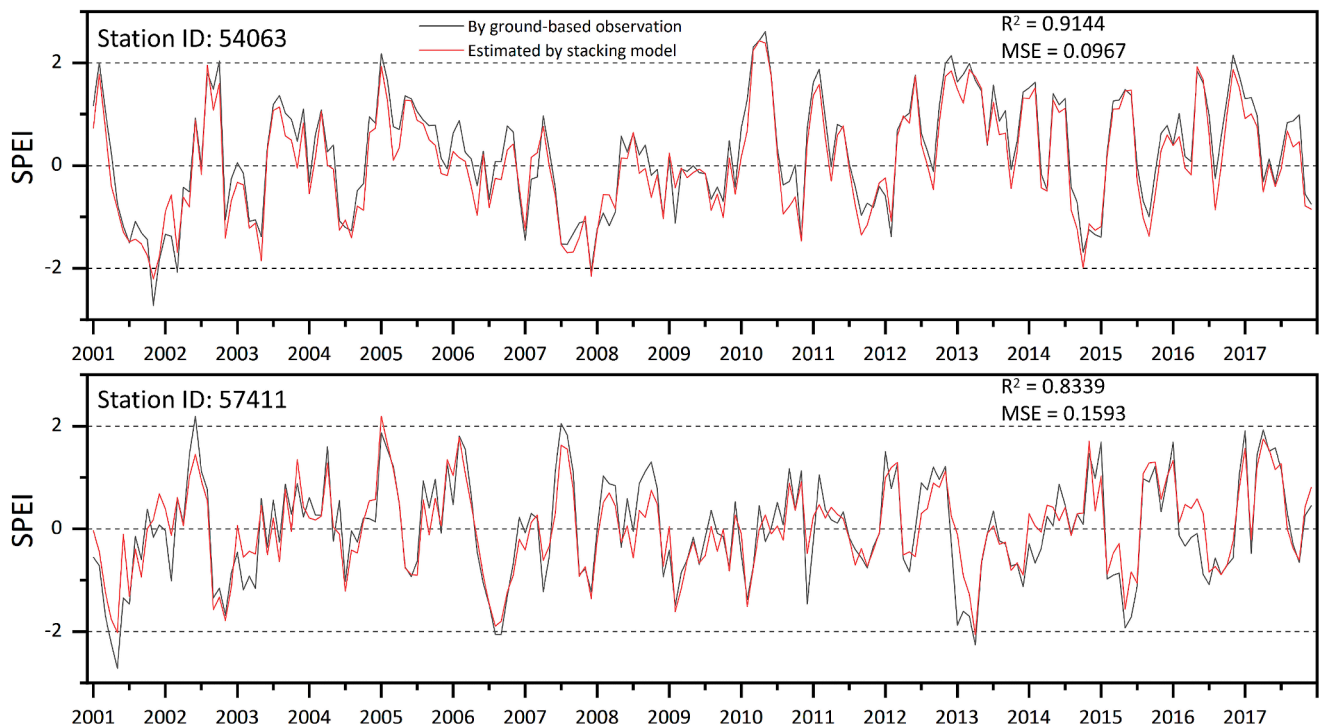


Fig. 8. Predicted SPEI-3 (2001–2017) based on the stacking models compared with the one derived from ground observed data at two stations. Station 54,063 is in the Northeast Plain, while station 57,411 is in the Sichuan Basin and Its Surrounding Areas.

of vegetation indices in drought estimation is relatively low, possibly because vegetation response to drought takes some time and different types of land cover have a large variation in sensitivity to drought (Yin et al., 2022). However, they still provide valuable information for

drought estimation. Therefore, the selection of remote sensing drought factors is crucial for accurate estimation and monitoring of drought and should be chosen according to specific application scenarios. We used a stacking model to estimate the drought conditions in China's cropping

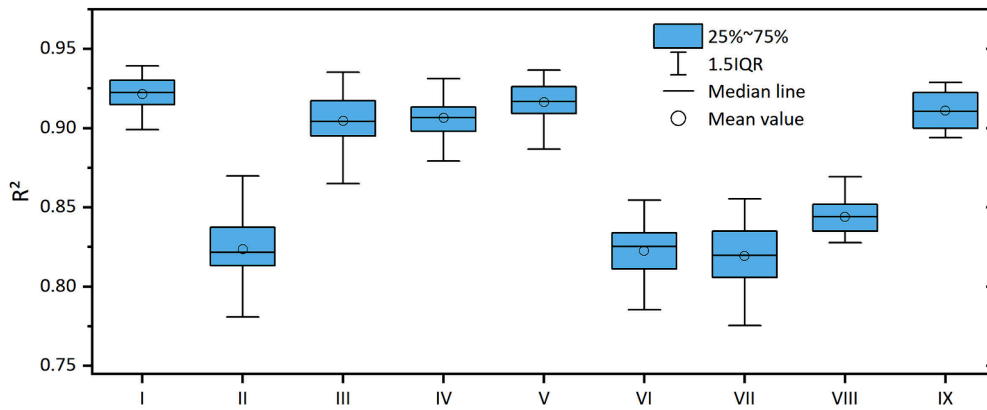


Fig. 9. Boxplot of R^2 values between observed and predicted SPEI-3 for the remaining stations in each subregion. I: Northeast Plain; II: The Semi-Arid and Arid Areas of Northern China; III: South China Region; IV: Huang-Huai-Hai Plain; V: Loess Plateau Region; VI: Qinghai-Tibet Plateau; VII: Sichuan Basin and Its Surrounding Areas; VIII: Yunnan-Guizhou Plateau; and IX: The Middle and Lower Reaches of the Yangtze River.

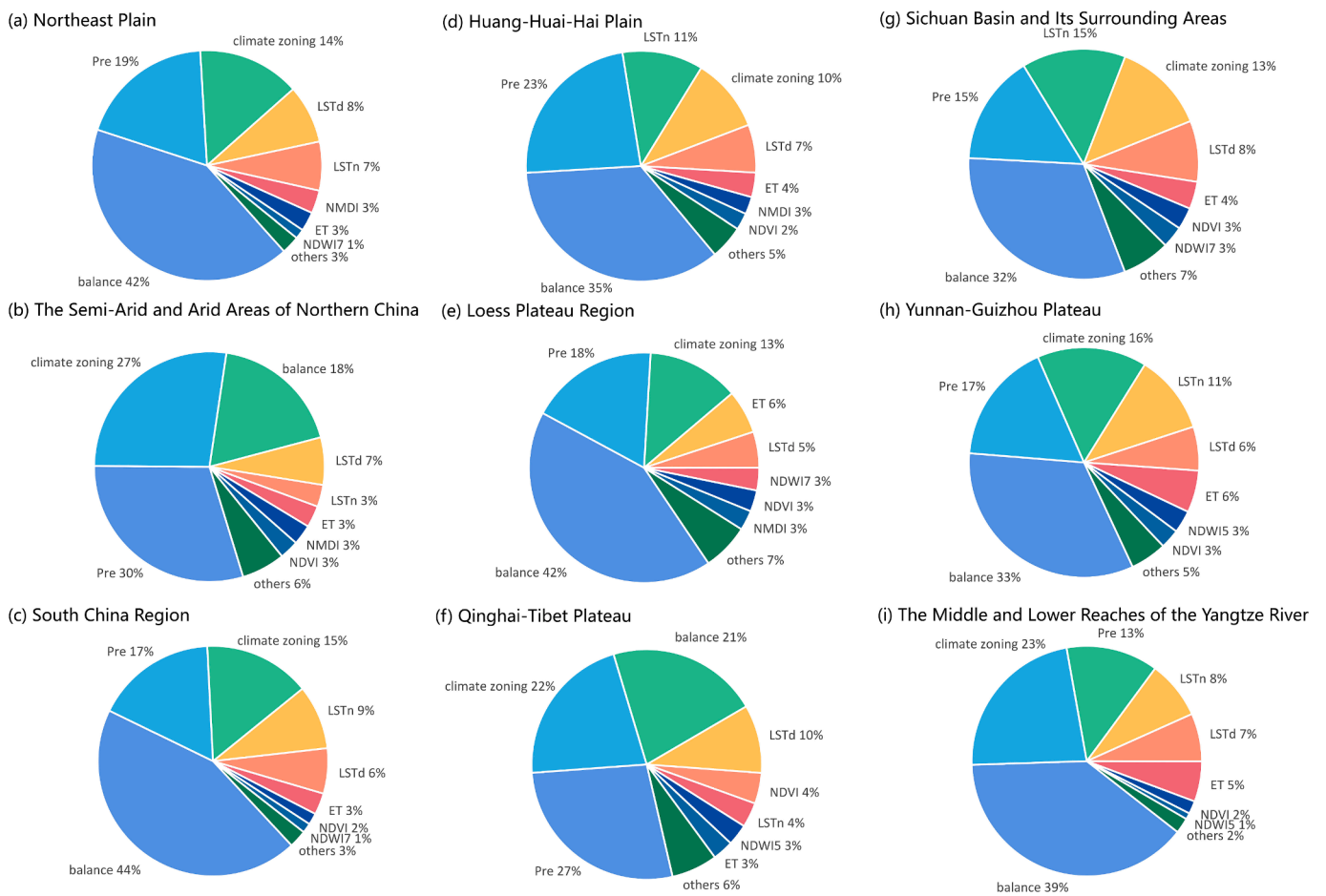


Fig. 10. Weighted relative importance for top nine remote sensing drought factors in the base model across 9 sub-regions. Climate zoning generated several features after dummy variable processing, and their values were summed when calculating the importance.

areas and found high consistency with drought maps generated by ground-truth GPCP and ERA5 data. Therefore, we believe that the remote sensing data fusion method based on machine learning ensemble models can produce reliable drought estimation. The improving capacity of multiple sensors observations helps us develop a deeper understanding of the mechanisms underlying drought, leading to more effective monitoring of drought (Jiao et al., 2021). The integration of machine learning and ensemble models for estimating drought using remote sensing data is more effective than simple models. The stacking

ensemble approach is only one of the model fusion and integration methods. Some related studies, such as those conducted by Alizadeh and Nikoo (2018) and Wang (2022), have also shown that model fusion-based methods can achieve better results in drought estimation. Our future work will consider using more data-fusion ensemble methods in drought prediction.

By utilizing our research method, we can offer valuable insights informing seasonal drought prevention and control policies in agricultural regions. Our findings provide evidence-based support, allowing

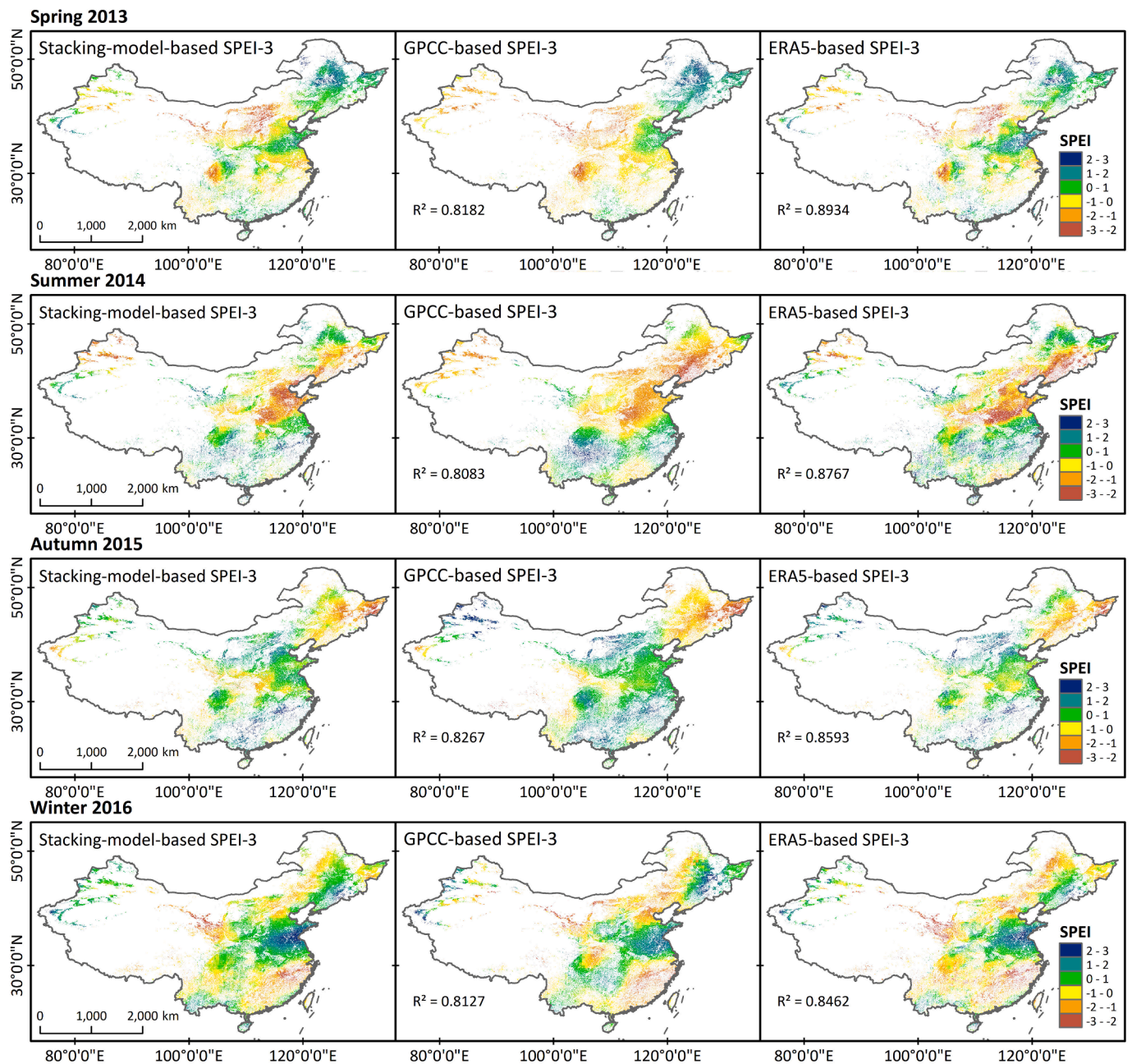


Fig. 11. Comparison between estimated SPEI based on the stacking model and GPCC/ERA5 products for the spring of 2013, summer of 2014, autumn of 2015, and winter of 2016 in China's cropping region. R^2 values were computed between the SPEI-3 estimated by the stacking model and those generated by the reanalysis datasets.

policy makers to make informed decisions regarding strategies and measures to mitigate the impacts of drought on agriculture. For example, as there is a significant correlation between SPEI and crop yield (Feng et al., 2019; Labudová et al., 2017; Mohammed et al., 2022; Wang et al., 2020), by analyzing seasonal drought maps based on our SPEI-3, we can help local farmers assess the sensitivity of different crops to drought by combining it with crop yield data. Meanwhile, by planting drought-resistant crops during dry seasons and high-value crops during wet seasons, our drought maps can optimize agricultural practices to mitigate the impacts of drought.

Though the accuracy in western China is slightly lower than that in eastern China, our research results are still significant because most of China's cropping areas are located in the east. However, we acknowledge that the default parameters of our machine learning methods and the use of 5 base models in the ensemble model may not necessarily

yield the best results. Nonetheless, these choices were made to strike a balance between computational complexity and the applicability of our approach. In future studies, we plan to enhance the applicability and accuracy of our research methods in regions with unique terrain and climate by using more appropriate remote sensing drought factors, denser site data, advanced model algorithms, and model integration methods. Additionally, we will focus more on predicting seasonal drought occurrences and assessing the likelihood of future drought events using precise drought estimation data to better inform decision-making and management.

5. Conclusions

This study compared the performance of 19 individual base models in estimating 3-month scale SPEI using fused remote sensing drought

factors from PERSIANN-CDR and MODIS satellites and China's climate zoning data processed by dummy variables across 9 agricultural regions in China. The best *meta*-model was selected among the 18 models to integrate the top 5 individual base models using stacking approach, and the performance of the stacking model was compared and evaluated. The results showed that CBR, ETR, XGB, LGBM, and RF were the top-performing base models in estimating drought, while CBR also performed well as the *meta*-model. The stacking model with CBR as the *meta*-model was utilized to estimate seasonal drought in spring 2013, summer 2014, autumn 2015, and winter 2016, and the results were compared with SPEI-3 produced by two reanalysis datasets, GPCC and ERA5. The stacking model integrated the strengths of the individual models and had good performance in estimating drought. When evaluating the performance of the stacking model, the accuracy in the eastern region of China was higher than that in the western region. When conducting large-scale drought estimation, the independent operation of each agricultural region and the incorporation of meteorological zoning data also improved the accuracy of the estimates.

Here we took China as a case study area. However, the methodological framework can be applied to other regions with vast areas and significant differences in terrain and climate, such as Europe and the United States. We suggest that more high-resolution remote sensing data should be used to provide more accurate and scientific guidance for seasonal drought at a local scale. Future research should further explore the use of more promising remote sensing drought factors, denser station data, advanced model algorithms, and model ensemble techniques for further enhancing the accuracy and reliability of drought monitoring.

CRediT authorship contribution statement

Xinlei Xu: Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Fangzheng Chen:** Resources, Writing – review & editing. **Bin Wang:** Writing – review & editing. **Matthew Tom Harrison:** Writing – review & editing. **Yong Chen:** Writing – review & editing. **Ke Liu:** Writing – review & editing. **Chenxia Zhang:** Writing – review & editing. **Meiqin Zhang:** Writing – review & editing. **Xueliang Zhang:** Writing – review & editing. **Puyu Feng:** Methodology, Supervision, Writing – review & editing, Funding acquisition. **Kelin Hu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the study are all publicly available online and links are provided in the text.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2024.131102>.

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